



Simon Says

Modules

- **simon_says_top**
- PRNG
- debouncer
- segment_display
- clk_divider

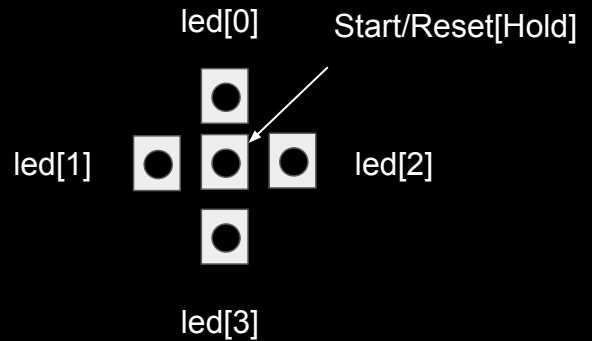


```
▼ ● 🚦 simon_says_top (simon_says_top.v) (8)  
  ● game_timer : clk_divider (clk_divider.v)  
  ● db0 : debouncer (debouncer.v)  
  ● db1 : debouncer (debouncer.v)  
  ● db2 : debouncer (debouncer.v)  
  ● db3 : debouncer (debouncer.v)  
  ● db_center : debouncer (debouncer.v)  
  ● game_randomizer : PRNG (PRNG.v)  
  ● screen : segment_display (segment_display.v)
```

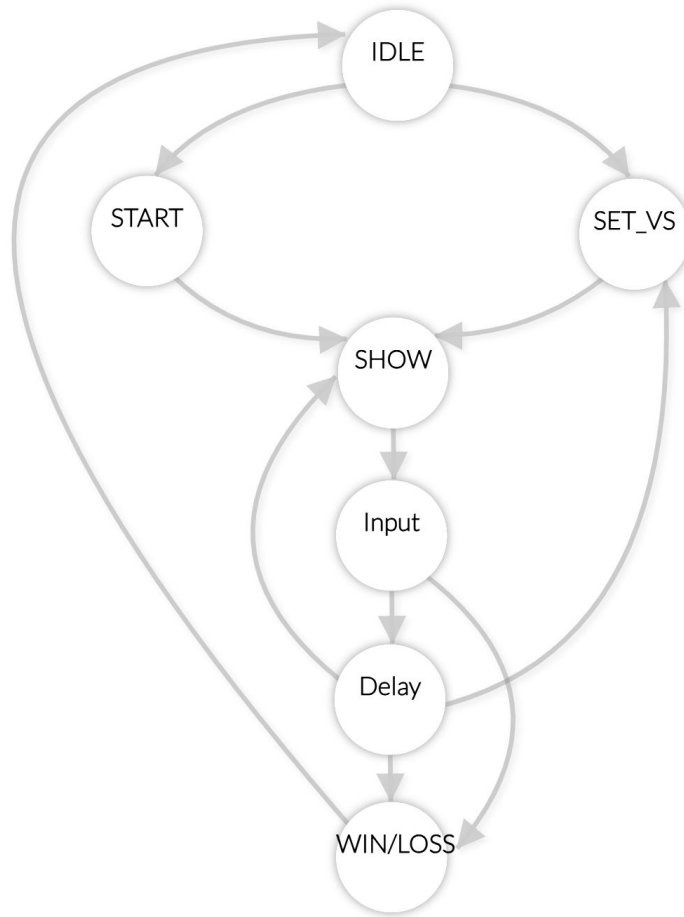
Unique Features



- Two different modes: Solo and Versus
- Delay after completing level
- Hold reset



FSM Logic



Challenges & Resolutions

- Memory Confusion: Versus mode distinguishing Player 1's moves and Player 2's moves
- Switch Mid-Game: Flipping the mode switch while a game was already running



THANK YOU

